

Applications of Queues

Q. Discuss the significance of Queues in computer science. Elaborate on various real-world and system-level applications where the Queue data structure is essential.

> **Definition to Remember**

> A **Queue** is a linear data structure that follows the First In First Out (FIFO) principle. It is used to store data elements in a specific order, where the element that is added first is the one that is removed first.

1. System-Level Applications (Operating Systems & Networks)

Application	Description
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2. Real-World & Algorithmic Applications

Application	Description
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3. Print Spooler Example Diagram

```
graph LR
    User[User] --> Queue[Queue]
    Queue --> Printer[Printer]
```

[User :> Queue :> Printer]

> **Must-Write Points (for 10 marks)**

> 1. Q

> **Quick Recall**

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